

**Trust Calibration in Teen-AI Decision Making: Evaluating Explainability, Peer Influence,
and Personalization in High-Risk Scenarios**

AP Research

April 30, 2025

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Literature Review

In recent years, the integration of Artificial Intelligence (AI) into decision-making processes has increased exponentially across various domains. AI systems are now routinely employed to assist humans in tasks ranging from low-stake scenarios, such as product recommendations to high-stakes scenarios, such as healthcare diagnostics and military operations. Numerous studies have highlighted how AI enhances decision-making efficiency and performance. However, the effectiveness of these systems is heavily influenced by human trust in AI, which can either facilitate or hinder collaborative performance. For instance, researchers from the Georgia Institute of Technology found that trust in AI was correlated with decision-making accuracy and performance in decision-making processes (Silva et al., 2022). These findings underscore the importance of calibrating trust in AI to ensure optimal collaboration and performance, preventing the pitfalls of overtrust and under-trust.

Both overtrust and under-trust represent two extremes of a trust spectrum that impair effective human collaboration. Overtrust occurs when users excessively rely on AI systems, often following recommendations blindly, even when flawed. For example, in a drone strike simulation, participants reversed their decisions when the AI provided conflicting guidance, leading to poorer task performance (Holbrook et al., 2024). Similarly, Okamura and Yamada (2020) developed a novel active trust calibration method that provided users with trust calibration cues (TCCs) such as an on-screen warning sign or warning message when overtrust was detected, significantly improving accuracy with these interventions. Conversely, under-trust occurs when users dismiss accurate AI recommendations in favor of personal judgment. In an inspection task, workers initially distrusted a new AI system, but as they adopted its use, they achieved greater efficiency (Schmid & Wiesche, 2023). These findings highlight that achieving a “Goldilocks” zone of trust—where trust is appropriately calibrated—is critical for optimizing decision-making performance (Allan et al., 2021).

In high-stakes scenarios, the implications of trust imbalances are magnified. Trust in AI tends to be lower in these scenarios as users are more cautious and skeptical when decisions carry significant stakes such as potential harm or loss (Ashoori & Weisz, 2019). This skepticism reflects the higher stakes associated with errors, making trust calibration particularly challenging yet quintessential. Studies suggest that trust calibration techniques are pivotal to calibrating trust. Among various methods, Explainable AI (XAI) features have demonstrated the most promise in adjusting user trust. These features aim to provide transparency by explaining how AI systems arrive at their decisions and give insights into how an AI system “thinks” as well as its possible limitations. For example, in a mushroom-picking simulation by Leichtmann et al. (2023), participants who used AI systems that lacked explanations tended to overtrust, whereas those who used an AI system with explanations developed a more adequate level of trust. Similarly, Weitz et al. (2019) found that participants rated AI systems that provided explanations to be more trustworthy, less deceptive, and were less wary of these systems.

The specific design of XAI features also plays a critical role in trust calibration. Kim and Song (2023) conducted a high-risk desert survival simulation (known as the Desert Survival Problem), providing insight into how framing the accuracy of AI outputs affects trust levels. They discovered that omitting framing altogether resulted in the most balanced trust compared to positively framed (e.g., “this AI is 80% accurate”) or negatively framed (e.g., “this AI is 20% inaccurate”) statements. The type of explanation provided can further influence user perceptions. Counterfactual explanations describing what could’ve happened (e.g., “he would’ve been over the limit if he drank on an empty stomach”) were judged as more helpful than causal explanations describing what did happen (e.g., “he was under the limit because he drank on an empty stomach”) (Celar & Byrne, 2023). Other studies have explored mechanisms for visualizing and better understanding AI decision-making processes. For instance, Gomez et al. (2023) created an AI to assist in bird classification. One version of the AI offered the user the three physical attributes that would be the most useful in identifying the bird, providing users with some insight into how the AI was coming up with its decision. This version was perceived to be more

knowledgeable and useful.

Trust in AI is not uniform but is influenced by individual differences, such as personality traits. The Big Five personality traits—openness, conscientiousness, extraversion, agreeableness, and neuroticism—affect trust levels differently. For example, individuals high in openness and agreeableness tend to perceive AI as more fair or trustworthy, whereas those high in neuroticism tend to perceive AI as less fair (Shulner-Tal et al., 2023). Furthermore, Gerlich (2024) used structural equation modeling (SEM) to discover that individuals with higher trust in AI had a higher distrust in humans, and vice versa. In addition, Gerlich (2024) found that customized content can significantly enhance trust in AI.

Despite the growing body of research on trust calibration, there is a notable gap in understanding how teens interact with AI systems. Unlike adults, teens often prioritize fairness, transparency, and social alignment when evaluating AI systems (Bochniarz et al., 2022; Higgs & Stornaiuolo, 2024). Teens tend to view AI as an artificial construct rather than a human-like system and are more distrustful of AI systems perceived as hostile or less governed by emotions (Bochniarz et al., 2022). Furthermore, Higgs and Stornaiuolo (2024) found that teens were nervous about AI technologies as they seemed to be “unfair” and take away from human creativity and ingenuity. Additional concerns include fears of overdependence and the spread of misinformation (Yu et al., 2024). In order to appropriately calibrate trust in teens, it’s important to address these unique concerns.

Teens’ decision-making processes differ significantly from adults, further necessitating tailored trust calibration techniques. Teens are more sensitive to rewards and exhibit higher risk-taking behaviors than adults (Paulsen et al., 2011; Sullivan-Toole et al., 2019). For instance, in an alien blaster simulation, teens showed a stronger response to rewards and a lower sensitivity to effort costs compared to adults (Sullivan-Toole et al., 2019). Similarly, when presented with risky and safe options with the same expected value, teens preferred the riskier, higher reward option (Paulsen et al., 2011). Emotional and social factors also play a significant role. In a study conducted by Albert et al. (2013), teens were more likely to engage in risk-taking behavior when

observed by a “peer” (participants were led to believe that a peer was watching), demonstrating heightened susceptibility to social influences. Similarly, Blankenstein et al. (2016) found that teens working with perceived peers were more risk-taking. This raises the possibility that teens may perceive AI systems as peer-like collaborators, further amplifying risk-taking behaviors. This dynamic makes peer influence a critical factor in understanding and calibrating trust in teen-AI interactions.

Existing studies largely focus on adult populations or general trust calibration techniques, leaving unanswered questions about how teens’ unique decision-making characteristics interact with AI systems. Addressing this gap is essential for advancing AI design to meet the needs of younger users. This study seeks to determine the best techniques to appropriately calibrate trust and improve decision-making abilities and performance for teens in high-risk scenarios. By specifically testing the effects of AI explainability, peer influence, and personality-tailored feedback, this research will contribute practical insights into trust calibration methods specifically designed for teens. By addressing teen concerns and tailoring methods to their decision-making process, this study aims to optimize both trust and performance in human-AI collaboration.

Method

This study employs a high-risk mushroom-picking simulation to investigate trust calibration in human-AI decision-making among teens. The selected method mirrors decision-making under uncertainty with tangible rewards and risks, providing an effective environment for testing the impacts of explainability, peer influence, and personality-tailored feedback on trust, accuracy, and decision-making speed. The design is modeled after prior research into human-AI collaboration, particularly the methodologies used by Leichtmann et al. (2023). Leichtmann et al. (2023) demonstrated the feasibility and effectiveness of using a mushroom-picking simulation to model real-world decision-making under risk, making it an ideal method for testing the proposed factors in this research study.

The simulation-based design was chosen because it offers a controlled and adaptable framework for systematically testing how experimental factors influence results and outcomes.

Real-world high-risk scenarios such as medical diagnostics or autonomous vehicle decision-making present significant ethical and logistical challenges, including risk to participants. The mushroom-picking simulation, on the other hand, provides a scaled-down and ethical alternative that retains the essential components of high-stakes decision-making.

The participants will be presented with 15 mushroom images along with AI-generated recommendations. These AI recommendations are based on real probabilities from a Convolutional Neural Network (CNN) trained for mushroom classification. For each image, participants will decide whether to classify a mushroom as edible or poisonous and then determine whether they would like to take it home. In total, participants will make 30 decisions—15 classifications and 15 take-home decisions.

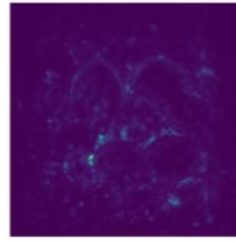
Participants will be randomly assigned to one of eight experimental conditions based on a $2 \times 2 \times 2$ factorial design, making sure to test each combination of the three factors (explainability, peer influence, and personality-tailored feedback) systematically. This particular experimental design allows for the exploration of both main effects and interactions between the factors themselves, providing nuanced insights. To manipulate explainability, some participants will be provided with visual explanations of the AI's classification using SmoothGrad saliency maps, which highlight the parts of the image that the AI model emphasized during its decision. For peer influence, some participants will see peer endorsements (e.g., “80% of users agreed with this choice”). Finally, for personality-tailored feedback, the AI's phrasing and feedback will adapt to participants' Big Five personality traits.

While exposing participants to all three factors at once risks information overload, the experimental design ensures that statistical analyses can mitigate these effects, allowing for accurate results. Figure 1 provides an example of the survey screen for a participant belonging to the group that has all three factors present.

Prior to the simulation, participants will complete a pre-task survey to determine their Big Five personality traits. The survey consists of 20 questions adapted from the International Personality Item Pool (IPIP), a validated measure of the Big Five personality traits. This

AP Research Experiment
Mushroom 1 of 15

Mushroom Image



A visualization of where the AI is focusing its attention

AI Recommendation: Edible: 40.49%, Poisonous: 59.51%.

A majority of your peers decided to not take this mushroom home.

Based on your personality test, you should look carefully before making your decision.

Figure 1

A sample participant screen for a participant belonging to the group with all three factors present

Mini-IPIP was chosen over the full-length 50-question IPIP version to minimize participant fatigue while maintaining sufficient accuracy. The abbreviated survey also allows for the subjects of the study, teens, to complete the method during their class time. Once the survey is complete, participants' scores will be calculated and used to tailor the AI's feedback in the simulation.

During the simulation, each image was accompanied by a series of four questions, including the classification decision, the take-home decision, and two Likert-scale items: "I trust this mushroom identification by the AI" and "I understand how the AI arrives at this mushroom classification." Figure 2 shows a sample of the questions participants were asked for each mushroom.

After completing the mushroom-picking simulation, participants will complete a post-task survey designed to measure their trust and comprehension of the AI. Once again, this will be modeled after the surveys that Leichtmann et al. (2023) provided in their study. In addition to survey data, the method will also collect five quantitative metrics to assess decision-making performance.

First, classification accuracy will measure how often participants correctly classify mushrooms as edible or poisonous. A correct classification would correspond to an edible

How would you classify this mushroom?

Would you take this mushroom home?

I trust this mushroom identification by the AI:

☐ Strongly Disagree
 ☐ Disagree
 ☐ Neutral
 ☐ Agree
 ☐ Strongly Agree

I understand how the AI arrives at this mushroom classification:

☐ Strongly Disagree
 ☐ Disagree
 ☐ Neutral
 ☐ Agree
 ☐ Strongly Agree

Figure 2

A sample questions screen

mushroom being classified as edible or a poisonous mushroom being classified as poisonous. Second, questionnaire accuracy will measure the correctness of the participants' responses to the post-task questionnaire. An average trust rating will be calculated by using the values from the "I trust this mushroom identification by the AI" five-point Likert scale. Similarly, the average comprehension ratings were calculated by using the values from the "I understand how the AI arrives at this mushroom classification" five-point Likert scale. Finally, the average response time will be measured to evaluate how quickly participants made their decisions under different conditions. Together, these metrics provide a comprehensive understanding of participants' decision-making processes and their reliance on the AI system.

However, the method does come with its limitations. The mushroom-picking simulation simulates consequences, but it can't fully mimic the high stakes or real-world pressure of actual high-risk scenarios. Another limitation is the reliance on surveys, which may be affected by participant inattentiveness when responding to survey questions. To address this, the surveys will employ "attention-check items," as recommended by Leichtmann et al. (2023). These are simply questions in which the answer is provided in the directions and can be used to screen out inattentive participants.

Despite these constraints, the simulation-based method remains the most appropriate design for this study. By integrating findings from prior research methods, this method offers a robust framework for exploring trust and performance in human-AI collaboration. By using

carefully chosen surveys and behavioral metrics, this design ensures both precision and relevance, providing insights into optimizing trust calibration and performance in AI systems designed for teens.

Results and Discussion

This study examined the effect of three experimental factors—AI explainability, peer influence, and personality-tailored feedback—on participants’ trust, decision-making abilities, and performance in a high-risk decision-making scenario. To best quantify these metrics, five dependent variables were measured and analyzed: average trust ratings, average comprehension ratings, average time spent, classification accuracy, and questionnaire accuracy.

To assess the effect of the independent variables on the five dependent variables, a Permutational Multivariate Analysis of Variance (PERMANOVA) was conducted. The PERMANOVA was selected over traditional parametric ANOVA tests due to violations of normality in experimental data. Additionally, the PERMANOVA allowed for the simultaneous assessment of multiple dependent variables, offering a robust framework for evaluating complex, multifactorial relationships across multiple variables.

The p -values obtained from the PERMANOVA were as follows: average trust rating ($p = 0.49$), average comprehension rating ($p = 0.082$), average time spent ($p = 0.182$), classification accuracy ($p = 0.375$), and questionnaire accuracy ($p = 0.744$). These results are summarized in Table 1. The null hypothesis is that the three independent variables have no effect on any of the five dependent variables that were measured. None of the factors had a statistically significant effect on any of the measured outcomes ($\alpha = 0.05$), so the null hypothesis fails to be rejected. The initial hypothesis is also incorrect.

Although none of the PERMANOVA tests yielded statistically significant results, some patterns were observed that merit further attention. The p -value for the average comprehension rating ($p = 0.082$) approached the significance level, suggesting that there may be subtle effects on comprehension worth exploring in future studies. Additionally, there also may be small effects on the average time spent per item. However, the overall lack of significant results across all five

Table 1*PERMANOVA Results*

Variable	<i>p</i> -value
Average Trust Rating	0.490
Average Understanding Rating	0.082
Average Time Spent	0.182
Classification Accuracy	0.375
Questionnaire Accuracy	0.744

Note. * $p < 0.05$.

dependent variables challenges the assumption that techniques designed and proven for adults will necessarily generalize to and be effective for teens.

Interestingly, task performance didn't improve with the addition of any of the independent variables. In fact, many participants consistently misclassified certain mushroom types regardless of group assignment, suggesting that trust alone may not be sufficient to override perceptual or cognitive biases in high-stakes tasks. Similarly, questionnaire accuracy, which assessed participants' comprehension of how the AI model worked, remained relatively flat across conditions, indicating that none of the techniques substantially improved participants' understanding of the system.

Taken together, these results suggest that teens may require alternative or simplified mechanisms for trust calibration in AI systems, and that direct use of adult-centered techniques may not be effective. The study's design enabled a robust comparison of multiple variables simultaneously, but the absence of statistically significant effects points toward the need for rethinking how teens engage with and interpret AI support in complex decisions.

Conclusion

The study provides new insight into how certain factors, particularly explainability, peer influence, and personality tailored feedback impact trust, decision making abilities, and performance for teen-AI systems in high-risk scenarios. The results demonstrated that none of these factors had a statistically significant trust on any of the metrics tested. This suggests that existing techniques designed and proven for adults may not directly translate to teens, highlighting the need for further teen-centered research in this area.

Several limitations should be considered when interpreting these findings. The first major limitation was the potential for an information overload. Since the study tested three different factors, certain groups of participants were presented with all three factors simultaneously. This meant that they had to process an additional image and two extra statements of text, leading to a potential information overload. This overload would cause participants to revert back to simplistic decision-making processes or arbitrary judgments not based on AI recommendations. Future research should explore ways to simplify how factors are presented in order to reduce information overload, perhaps by introducing them in stages rather than all at once.

The second limitation was the participant population. The study was conducted with high school students from a STEM focused magnet school, a group that often has a very unique experience and understanding of AI compared to most high schoolers their age. Their pre-existing understanding and attitudes towards AI could've influenced how they interacted with the AI system, possibly reducing the intended effect of the experimental conditions. Future research should recruit participants from a broader range of educational and technological backgrounds as well as AI familiarity to ensure that findings reflect a more representative spectrum of teen experiences.

Despite these limitations, the findings of this study carry several important real-world implications. As AI tools continue to enter domains such as education, health, and safety, it is essential to understand how different age groups, especially teens, interact with these systems. Understanding the interactions with AI is critical for designing effective decision-support tools.

However, designing decision-support tools for teens requires more than adapting adult-centered techniques. It calls for a deeper understanding of the cognitive, social, and developmental characteristics that shape teen-AI interactions. For instance, developers might consider incorporating more intuitive feedback mechanisms or gradually layering trust-building features to avoid overwhelming younger users.

The study also suggests several promising directions for future research. The first future direction that should be taken is to explore the use of a longitudinal study. This would examine how repeated interactions with AI influence trust and decision making performance over time since current literature doesn't explore interactions with AI over time. This type of study would be even more important now as the role of AI in our daily lives is rapidly increasing and will continue to do so over time.

The next future direction that should be taken is to find the factors that do produce a significant effect on teens and learn why these factors are more effective for teens compared to the three factors that were explored in this study. Understanding what mechanisms actually shape teen trust is essential for designing AI systems that are both effective and well-received.

As AI continues to play a growing role in decision-making, it is crucial to understand how different age groups, particularly teens, interact with AI in decision making processes. Overall, this study challenges assumptions about AI trust-building strategies, showing that common techniques that work for adults do not necessarily translate to younger users. These findings underscore the need for more specific approaches to working alongside AI. By continuing to explore how AI systems can be designed to align with teen cognitive processes and decision-making tendencies, future research can contribute to the development of AI systems that are not only more effective but also more trustworthy for the next generation of users.

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[Manuscript submitted for publication, to appear in the IEEE Symposium on Security and Privacy (S&P 2025)].